Spectral Rendering and Dispersion in a Path Tracer

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1 Overview

Standard path tracers model light as a single RGB, ignoring that physical light is a distribution over wavelengths. This omission makes it impossible to render dispersive phenomena such as prismatic rainbows and chromatic aberration in glass. For PA4, we extended the CS248A path tracer with a full spectral pipeline: each ray carries a sampled wavelength, the glass BSDF uses a wavelength-dependent index of refraction derived from the Cauchy equation, and accumulated radiance is converted to sRGB through the CIE 1931 color matching functions. The result is a physically correct dispersion rendered entirely inside the existing Slang shader framework.

2 Implementation

Wavelength Sampling

We added a `float wavelength` field to the `Ray` struct in `math/ray.slang`. At the start of each path the renderer samples a wavelength uniformly from the visible spectrum $[380, 700]$ nm:

```
wavelength = 380.0 + rng.next_1d() * 320.0;
```

The wavelength is preserved through all bounces by copying it onto every child ray, ensuring a single photon path is monochromatic end-to-end, which satisfies the Monte Carlo integration requirement.

Cauchy Dispersion Equation

The wavelength-dependent index of refraction is computed with the empirical Cauchy formula:

$$n(\lambda) = A + \frac{B}{\lambda^2}$$

We use flint-glass constants $A = 1.7$, $B = 6200$ nm^2. At the sodium D-line (589 nm) this gives us $n \approx 1.718$, matching tabulated data. The dispersion $n(400\,\text{nm}) - n(700\,\text{nm}) \approx 0.026$ produces clear visible color separation. The IOR replaces the previously fixed constant in `GlassBRDF.sample()`. For more vivid artistic renders B can be increased to 12,000 nm^2.

CIE 1931 Color Matching

To convert each monochromatic sample to sRGB we used the CIE 1931 2-degree standard observer. The three color-matching functions $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$ are tabulated at 81 points from 380–780 nm in 5 nm steps inside `utils/spectral.slang`, with linear interpolation between samples. The XYZ triplet is converted to linear sRGB via the IEC 61966-2-1 D65 matrix:

$$
\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 3.2406 & -1.5372 & -0.4986 \\ -0.9689 & 1.8758 & 0.0415 \\ 0.0557 & -0.2040 & 1.0570 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}
$$

Negative sRGB values are left unclamped during accumulation, as they cancel correctly in the Monte Carlo sum and clamping occurs only at tone-mapping time.

Spectral Accumulation

For each pixel the path tracer fires N samples each with an independently drawn wavelength. The shading luminance is multiplied by the spectral sRGB color of that wavelength and accumulated:

$$
\text{pixel\_rgb} \approx \frac{1}{N} \sum_i \text{sRGB}(\lambda_i) \cdot L(\lambda_i)
$$

A `spectralMode` flag in `RendererUniform` allows changing between the spectral and original RGB paths with no need to recompile.

3 Scenes and Results

Triangular Glass Prism

A triangular prism is constructed with a 40° apex angle and illuminated by a narrow rectangular area light. Light entering one angled face refracts at a wavelength-dependent angle and exits through as rainbows. Rendered at 2048–4096 SPP with path depth 6. The Cauchy B constant was tuned to 12,000 nm^2 for vivid color separation.

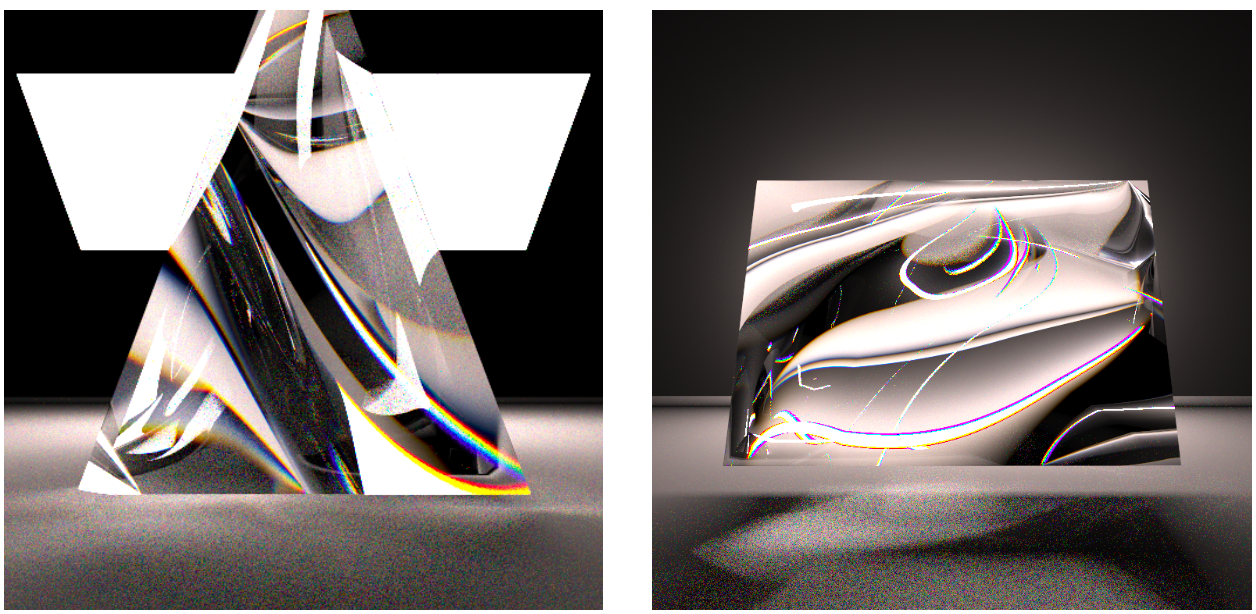

Figure 1: Left: glass prism with spectral dispersion and rainbow fringing at the base. Right: prism rainbow at higher SPP showing clean color separation.

Prism Caustics on Ground Plane

With the prism elevated above a Lambertian ground plane, dispersed rays refract twice then reemerges as rainbow caustics below. Capturing this required a glass bounce loop allowing up to ten refraction events per ray. The caustic colors are a direct consequence of wavelength-dependent bending: red reemerges at a different position than blue.

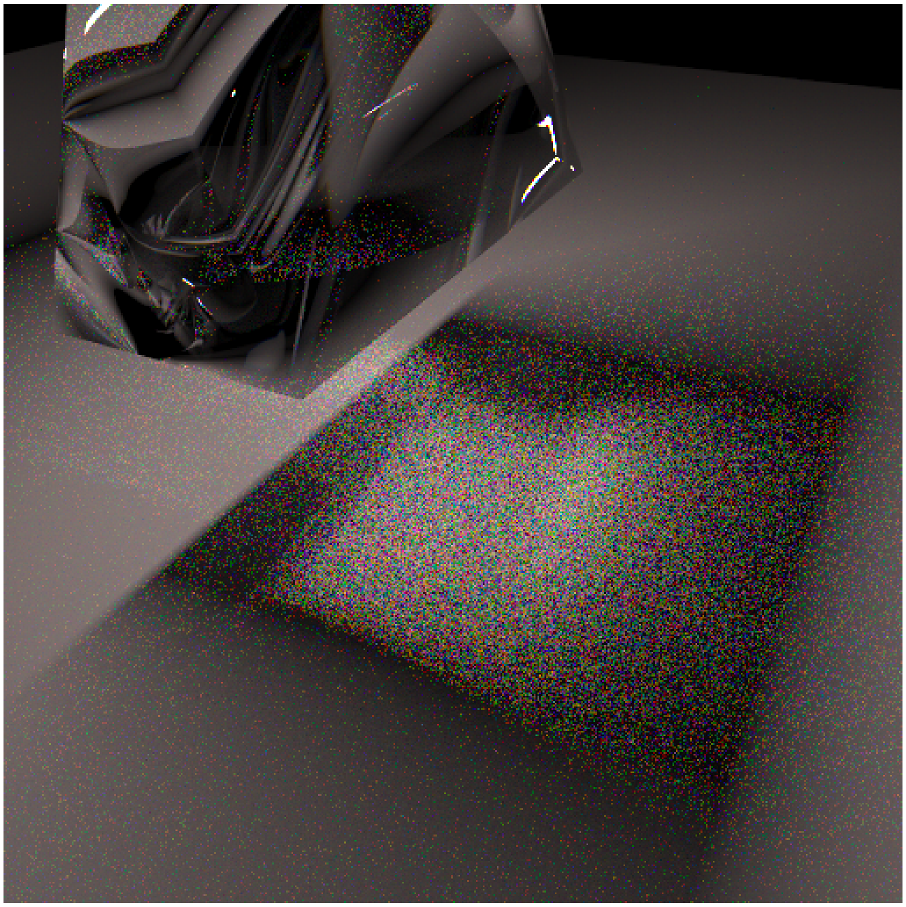

Figure 2: Rainbow caustics on the ground plane below the prism.

Glass Sphere — Chromatic Aberration

A smooth UV sphere with a glass BRDF is lit by a point light and a rectangular area light. Chromatic aberration appears as colored fringing along the silhouette edge where refracted rays of different wavelengths exit at slightly different angles. Rendered at 1024 SPP with path depth 4.

4 Discussion

The spectral extension adds minimal overhead: one Cauchy evaluation (two floating-point operations) and one CIE table lookup (two lerps) per ray. Convergence requires more samples than RGB rendering because each sample contributes only one wavelength's worth of information. In practice 1024 SPP produces clean results for diffuse surfaces while 2048–4096 SPP is needed for clean caustics through glass.

A natural extension beyond this is hero wavelength sampling, which draws multiple correlated wavelengths per path to reduce variance at no additional path-tracing cost. The Fresnel term could

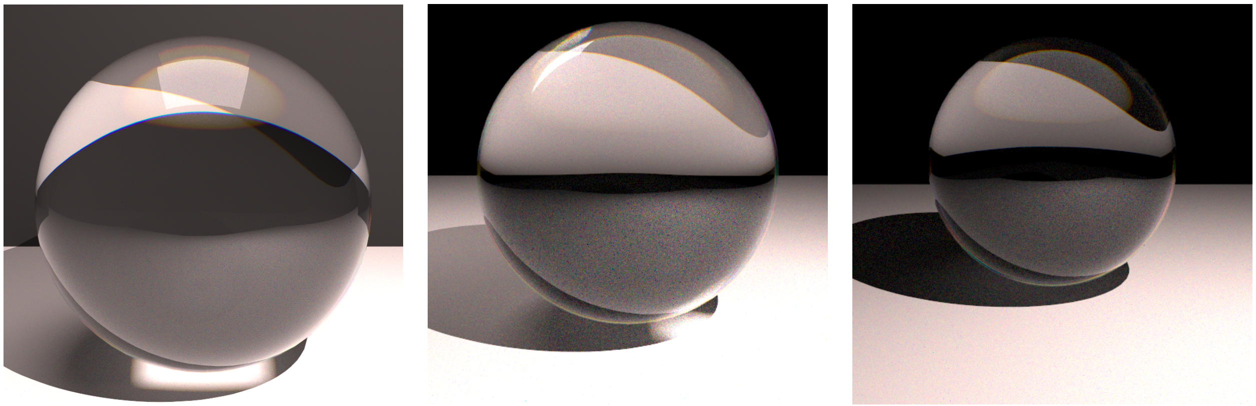

Figure 3: Glass sphere renders showing chromatic aberration fringing at the silhouette under different lighting setups.

also be made fully spectral in the same way as the refraction.

References

CIE 15:2004, *Colorimetry*, 3rd ed., Vienna.
Pharr, Jakob & Humphreys (2023). *Physically Based Rendering*, 4th ed. MIT Press.
Cauchy, A.-L. (1836). Sur la dispersion de la lumière.
IEC 61966-2-1:1999, *Default RGB color space — sRGB*.